PROTOMOTA

Brad Dunlap

Technology leader in AI, product and engineering

Bradenton, FL brad@protomota.com braddunlap.info linkedin.com/in/bradley-dunlap

1 Summary

Technology leader with 20+ years automating business and technology work, from hand-written scripts, test-driven development and CI/CD pipelines at RIIS to AI agents built with Claude and Codex today. Focuses on signal over noise and roots out performance theater: work that checks boxes but doesn't move the needle. At NTT DATA, leads the team behind a Fortune 500 AI pilot. Now, with AI as a force multiplier, builds apps, robots and AI agent systems solo through Protomota Lab.

Table 1. Key figures

Professional experience	20+ years
Overruns during scoping program	0 projects
Apps built solo, live on the App Store	2
Self-published websites and web apps	6
YouTube channels	2
Automotive brand apps led	7
Shopping center apps directed	15

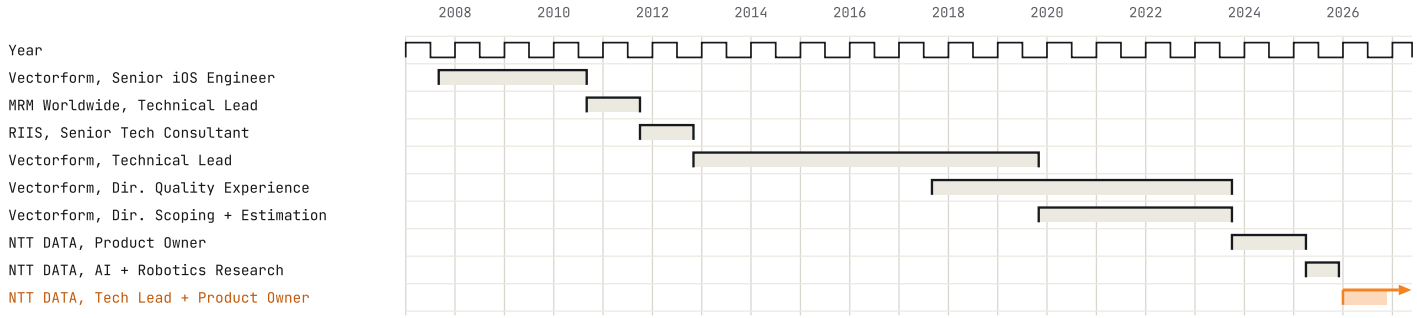


Fig. 1. Employment timing diagram, 2007 to present. Overlapping traces are concurrent leadership roles at Vectorform.

2 Professional experience

NTT DATA

Digital Solution Consultant Associate Director — Oct 2023 – Present

Technical Lead and Product Owner — Jan 2026 – Present

- Lead a small multidisciplinary engineering team building VR, voice and AI-assisted automation applications for a Fortune 500 client's confidential workflow initiative, now in pilot.
- Lead Azure, Next.js and Unity architecture, including Infrastructure as Code (IaC) and CI/CD.
- Build AI and voice capabilities: speech recognition, text-to-speech, conversational AI and voice commands.

Research and Development in AI and Robotics Simulation — Apr 2025 – Dec 2025

- Built agentic systems with Claude and Codex to coordinate technical, research and delivery work.
- Simulated robots in NVIDIA Omniverse and Isaac Sim and trained them in NVIDIA Isaac Lab; built ROS 2 workflows connecting hardware behavior and software-driven test environments.

Product Owner, Enterprise Mobile Application — Oct 2023 – Apr 2025

- Owned a multi-year iOS and Android application for a top U.S. manufacturing company, translating product direction into epics, user stories and acceptance criteria for a 20+ person team in Azure DevOps.

Protomota Lab

Independent work outside professional employment

Founder and Independent Developer — protomota.com

- Built Jotley solo, from concept through all development, marketing and training materials; live on the App Store.
- Built PulseCanvas Studio solo the same way; live on the App Store for Mac, iPhone, iPad and Apple Vision Pro.
- Build AI agents with Claude and Codex, including a dashboard that automates lab workflows.
- Built REST APIs for personal apps and automated their environment setup and releases.
- Run airesourcepro.com and two YouTube channels: free tutorials and videos for learning AI by building with it.
- Build and program robots with ROS 2, simulate them in NVIDIA Isaac Sim and train them in NVIDIA Isaac Lab.
- Design robot parts and 3D models in Plasticity CAD and Blender, and 3D-print them.

Vectorform

Director of Scoping and Estimation
Nov 2019 – Oct 2023

- Built and led a scoping and estimation process that took the company from several overrun projects per year to zero during program ownership.
- Improved estimate accuracy and profitability by aligning scope, staffing, feasibility and risk with Sales and Delivery; introduced analytics comparing actuals against estimates.

Director of Quality Experience
Sep 2017 – Oct 2023

- Led a small QA team and brought TDD and automated UI testing from RIIS to Vectorform.

Technical Lead, Mobile Application Engineer
Nov 2012 – Nov 2019

- Led the full lifecycle of seven Stellantis/FCA apps: Mopar, Chrysler, Dodge, Jeep, Ram, Alfa Romeo, Fiat.
- Led the DTE Energy Insight App and real-time Energy Meter integration from concept through launch, supported FCC approval, and contributed technical and product expertise to its spin-off into a new company, Powerley.
- Directed 15 Taubman shopping center apps with Bluetooth and Wi-Fi indoor navigation. Automated environment setup and CI/CD releases for mobile projects.

RIIS

Senior Tech Consultant, Mobile Strategy and App Development
Oct 2011 – Nov 2012

- Wrote scripts to provision and configure development and test environments on a small team.
- Directed iOS and Android apps and introduced test-driven development, Jenkins CI/CD pipelines and automated UI testing.
- Reverse engineered and penetration tested clients' iOS and Android apps to help harden their security.

MRM Worldwide

Technical Lead, Front End
Sep 2010 – Oct 2011

- Led front-end development for Chevrolet.com's annual model refresh, defining architecture and coding standards.

Vectorform

Senior iOS Engineer
Sep 2007 – Sep 2010

- Founded Vectorform's mobile practice as its first iOS developer; led apps featured on the Apple App Store.

3 Technical skills

Area	Capabilities
AI and automation	AI engineering, Model Context Protocol (MCP), prompt engineering, AI agents, LLM fine-tuning, end-to-end workflow automation, Claude, Codex, Cursor
Product engineering	React, Next.js, Python, REST API development (personal apps), databases, Swift, SwiftUI, iOS, Android, Unity, virtual reality (VR), voice technologies, Internet of Things (IoT), technical design
Robotics and hardware	NVIDIA Omniverse, Isaac Sim, Isaac Lab, ROS 2, robot assembly and programming, electronics, 3D printing, Blender, Plasticity CAD
Platform and delivery	Infrastructure as Code (IaC), DevOps, CI/CD pipelines, GitHub Actions, Jenkins, Azure DevOps, Azure, AWS, Cloudflare, Linux administration, Docker, Proxmox, JIRA, Fastlane, release automation
Testing and security	Playwright, mobile UI automation, BrowserStack, test strategy, QA process design, mobile penetration testing, reverse engineering, vulnerability analysis

4 Education

Bachelor of Fine Arts (BFA) in Graphic Design, Eastern Michigan University

5 Certifications

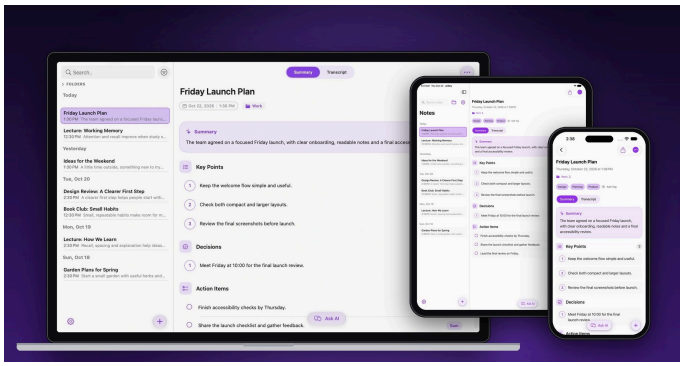
- Google IT Automation with Python Professional Certificate
- NTT Gen AI White, Yellow and Green Belt
- NTT DATA Agile and Scrum Internal Certification
- GitHub Copilot Certified
- Google AI Essentials

6 Industries

Manufacturing, utilities and energy, automotive, retail and shopping centers, IoT and connected hardware, sweepstakes and promotions, entertainment and consumer brands

7 Selected work

My client work is confidential, so these are my own builds from Protomota Lab.



Jotley

Fig. 2

Private AI notes for Mac, iPhone and iPad. Transcribes speech live, then summarizes and answers questions on the device with Apple Intelligence.

Built solo: concept, design, development, marketing and training materials. On the App Store.

`Swift, SwiftUI, Apple Intelligence, iCloud`

`jotley.ai`



PulseCanvas Studio

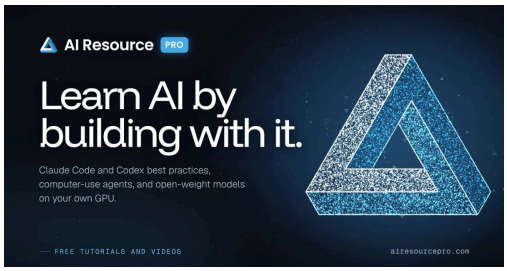
Fig. 3

Music visualizer and video maker for Mac, iPhone, iPad and Apple Vision Pro. Finds the beat, splits the song into sections and plays native visualizers and MilkDrop effects.

Built solo: concept, design, development, marketing and training materials. On the App Store.

`Swift, SwiftUI, Metal, export up to 4K at 60 fps`

`pulsecanvas.si`



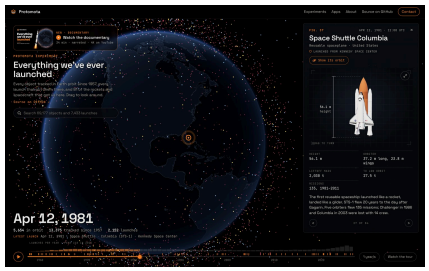
AI Resource Pro

Fig. 4

Free tutorials and videos for learning AI by building with it: Claude and Codex best practices, computer-use agents and open-weight models on your own GPU.

`airesourcepro.com`

`youtube.com/@airesourcepro`



Everything we've launched

Fig. 5

Interactive 3D simulation of every object tracked in Earth orbit since Sputnik, appearing launch by launch on a 3D Earth. Stop at milestone ships, search any object, and see how much of it is junk.

`protomota.com/experiments`

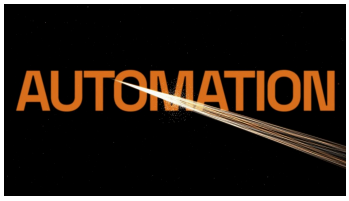


iPhone Duo teardown

Fig. 6

A folding iPhone rebuilt in code. Take it apart in repair order and tap any part to see where the numbers came from.

`protomota.com/experiments`



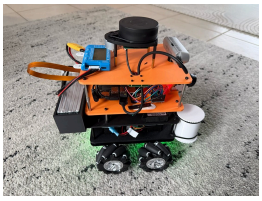
AI video and motion

Fig. 7

Clips, short films, titles and visuals that move to music.

`protomota.com/motion`

`youtube.com/@protomota`

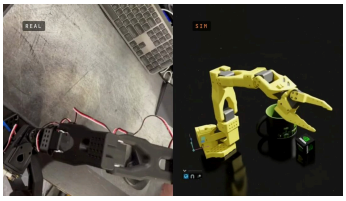


Project Ogre

Fig. 8

Mecanum-drive robot on a Jetson Orin Nano with SLAM and Nav2 autonomous navigation, trained in Isaac Lab.

`newsletter.protomota.com`



SO-ARM 101 digital twin

Fig. 9

A leader arm drives a 6-axis arm in Isaac Sim in real time.

`Build log`



JetBot Orin

Fig. 10

The NVIDIA JetBot, rebuilt for the Jetson Orin Nano with a custom chassis.

`Build log`

More work protomota.com github.com/protomota newsletter.protomota.com x.com/protomotalab